

Differentiable functions – 4

Taylor's formula: remainders and estimates of derivatives

Homework

1 сентября 2026 года

Theoretical minimum. Taylor's formula with the Lagrange, Cauchy and integral remainders; Peano's form and its use in computing limits; the Landau and Kolmogorov inequalities between $\|f\|$, $\|f'\|$ and $\|f''\|$; expansions at the endpoints of an interval; Bernstein's and Markov's inequalities for polynomials as the finite-dimensional analogue.

1. Compute $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$ and $\lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin x}{x^2}$ by Taylor expansion, and explain why l'Hôpital is the slower route here.
2. Let $f \in C^3$ with $|f| \leq 1$ and $|f'''| \leq 1$ on $\mathbb{R}$. Prove that $|f'| \leq C$ for an absolute constant C , and find the best constant you can.
3. Prove Kolmogorov's inequality $\|f'\|^2 \leq \|f\| \|f''\| \cdot c$ in the periodic case with the sharp constant, using Fourier series instead of Taylor.
4. Let $f \in C^{n+1}[a, b]$ vanish at $n + 1$ distinct points. Prove that $\|f\|_\infty \leq \frac{\|f^{(n+1)}\|_\infty}{(n+1)!} \max_x \prod_k |x - x_k|$, and deduce the error estimate for Lagrange interpolation.
5. Prove Markov's inequality $\|p'\|_\infty \leq n^2 \|p\|_\infty$ on $[-1, 1]$ for polynomials of degree n , and check that equality holds for the Chebyshev polynomial.
6. Let $f \in C^2[0, \infty)$ with $f(x) \rightarrow 0$ and f'' bounded. Prove that $f'(x) \rightarrow 0$.
7. Prove that if $f \in C^\infty(\mathbb{R})$ and $|f^{(n)}(x)| \leq n!$ for all n and all $|x| \leq 1$, then f is real-analytic on $(-\frac{1}{2}, \frac{1}{2})$, and give an example showing that the bound $n!$ cannot be replaced by $(n!)^2$.

8. Prove the second-order mean value estimate $|f(x) - f(a) - f'(a)(x - a)| \leq \frac{\|f''\|}{2}(x - a)^2$ and use it to show that a convex C^2 function with bounded second derivative grows at most quadratically.
9. Let $f \in C^2[0, 1]$ with $f(0) = f'(0) = 0$ and $f(1) = 1$. Prove that $\|f''\|_\infty \geq 2$ and describe the extremal functions.
10. Prove the Bernstein inequality $\|T'\|_\infty \leq n\|T\|_\infty$ for trigonometric polynomials of degree n and compare it with Markov's inequality for algebraic ones; explain where the difference between n and n^2 comes from.